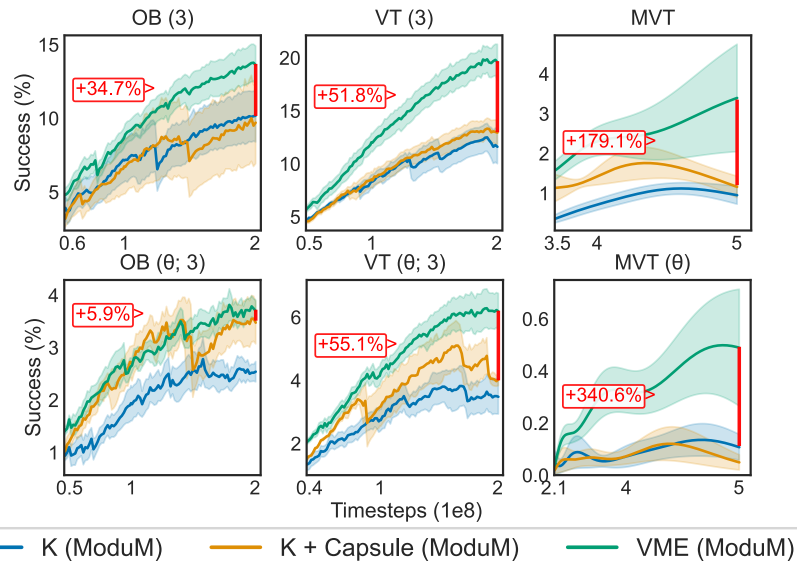

Figure 9: **Mesh-UNIM L Benchmark (Harder):** We observe that the tasks based on Mesh-UNIM L lead to much more unstable training curves, and a generalized loss of performance compared to equivalent UNIM L tasks. VME's context vectors improve performance in all tasks. We observe that VME provides much higher improvements in MVT, where K (ModuM) method seems to fail to learn much useful behaviour. This can be explained by the higher amount of rich robot-object-terrain contacts necessary to have traction over solving the task. The red labels and lines correspond to the improvement delta between `VME` and the next-best performer.

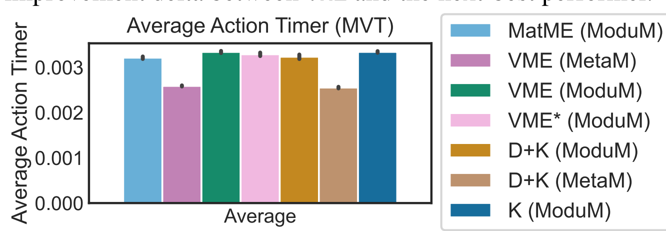

Figure 10: **Time needed to generate an action** (averaged over batch size and all training timesteps). We notice that all methods take approximately 0.003 seconds to generate an action, with only small differences. The timer was set up to capture the inference time for the exact line in the code where the model was called.

The UNIM L dataset was select for its great diversity of morphological graphs, unparalleled by any other alternative. This, however, limited our experiments to robots composed of primitive shapes. Yet, one of the core features of VME is its ability to ingest and describe any input shape, including arbitrary meshes (governed by voxel grid size). To showcase this feature of our proposed method, we have develop a suitable alternative. We constructed a synthetic version of UNIM L wherein we randomly sampled meshes from the Mujoco Menagerie dataset. For each limb, for each UNIM L robot, we sampled a random mesh from Mujoco Menagerie, and we (greedily) resized, scaled, and sheared the mesh to fit the limb. We then trained three variants of ModuM on this 'Mesh-UNIM L' dataset: VME (ModuM), D (ModuM), and D+Capsule (ModuM). The first two correspond to baselines used throughout this paper. To provide a fair comparison to a strategy commonly used in RL for locomotion tasks, wherein capsule approximations are used to represent meshes, we also showcase D+Capsule. In other words, the context vector in D+Capsule (ModuM) on Mesh-UNIM L corresponds to the D+K (ModuM) context on UNIM L. Due to the very high computational cost of training controllers on Mesh-UNIM L (we found simulation to be much costlier than UNIM L), we only used the three harder tasks from UNIM L: OB, VT, and MVT. The results are shown in Figure 9.

In Figure 10, we show that all methods take essentially the same time to generate actions. ny differences can be attributed to random noise or hardware differences.